Chapter 4. Differentiable Functions

Derivatives, mean value theorems, L'Hôpital's rule, Darboux's theorem, Taylor's theorem

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4.1 Differentiability

Motivation. The derivative is the best linear approximation to a function near a point. Its algebraic rules (product, quotient, chain) follow from writing $f(x+h) = f(x) + f'(x)h + o(h)$.

Let I be an interval. $f : I \to \mathbb{R}$ is **differentiable** at $x_0 \in I$ if $f'(x_0) = \lim_{h\to 0} \frac{f(x_0+h)-f(x_0)}{h}$ exists (one-sided at an end point).

Proposition 4.1 (4.1.1)

For $f : (a, b) \to \mathbb{R}$ and $x_0 \in (a, b)$ the following are equivalent:
(a) f is differentiable at x_0;
(b) there are $\lambda \in \mathbb{R}$ and a function ϕ with $f(x_0+h)-f(x_0) = \lambda h+|h|\phi(h)$ and $\lim_{h\to 0} \phi(h) = 0$.
Then $\lambda = f'(x_0)$.

Proof. (a)⇒(b): put $\lambda = f'(x_0)$ and $\phi(h) = (f(x_0 + h) - f(x_0) - \lambda h)/|h|$ for $h \neq 0$, $\phi(0) = 0$.
(b)⇒(a): $\frac{f(x_0+h)-f(x_0)}{h} = \lambda \pm \phi(h) \to \lambda$. ■

The linear map $h \mapsto \lambda h$ is the best linear approximation of $f(x_0 + h) - f(x_0)$; this is the form of the derivative that generalizes to several variables.

Proposition 4.2 (4.1.2)

If f is differentiable at x_0, it is continuous at x_0.

Proof. $f(x_0 + h) - f(x_0) = \lambda h + |h|\phi(h) \to 0$. ■

Proposition 4.3 (4.1.3)

If f, g are differentiable at x_0, so are $f+g$, fg, and f/g (if $g(x_0) \neq 0$), with $(f+g)' = f'+g'$, $(fg)' = f'g + fg'$, $(f/g)' = (f'g - fg')/g^2$ at x_0.

Proof. For the product, $\frac{f(x_0+h)g(x_0+h)-f(x_0)g(x_0)}{h} = \frac{f(x_0+h)-f(x_0)}{h} g(x_0+h) + f(x_0)\frac{g(x_0+h)-g(x_0)}{h}$, and $g(x_0 + h) \to g(x_0)$ by Proposition 4.2. For the quotient, first $\frac{1/g(x_0+h)-1/g(x_0)}{h} = -\frac{g(x_0+h)-g(x_0)}{h} \cdot \frac{1}{g(x_0+h)g(x_0)} \to -g'/g^2$, then use the product rule. ■

Proposition 4.4 (4.1.4, chain rule)

If $f : I \to J$ is differentiable at x_0 and $g : J \to \mathbb{R}$ is differentiable at $y_0 = f(x_0)$, then $g \circ f$ is differentiable at x_0 and $(g \circ f)'(x_0) = g'(y_0)f'(x_0)$.

Proof. Write $g(y_0 + k) - g(y_0) = k(g'(y_0) + \psi(k))$ with $\psi(k) \to 0 = \psi(0)$. With $k = f(x_0 + h) - f(x_0) \to 0$,

$$\frac{g(f(x_0 + h)) - g(f(x_0))}{h} = \frac{k}{h}\left(g'(y_0) + \psi(k)\right) \to f'(x_0)g'(y_0).$$

■

Example 4.1

$f(x) = x^2 \sin(1/x)$, $f(0) = 0$, is differentiable everywhere with $f'(0) = \lim h \sin(1/h) = 0$, but $f'(x) = 2x \sin(1/x) - \cos(1/x)$ is not continuous at 0. $|x|$ is continuous but not differentiable at 0.

4.2 Mean value theorems

Motivation. A derivative is local information, but we usually want global conclusions: $f' \geq 0$ implies f is increasing, $f' = 0$ implies f is constant. The mean value theorem converts one into the other, and its generalizations give L'Hôpital's rule.

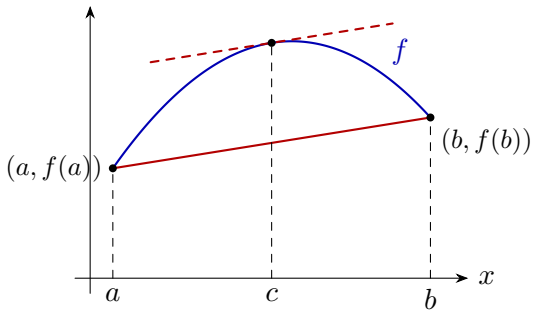


Figure. *The mean value theorem: at some $c \in (a,b)$ the tangent line (dashed) is parallel to the secant, i.e. $f'(c) = (f(b) - f(a))/(b-a)$.*

If f has a local maximum or minimum at an interior point c and is differentiable there, then $f'(c) = 0$: the difference quotients have opposite signs on the two sides.

Lemma 4.5 (4.2.1, Rolle)

If $f : [a,b] \to \mathbb{R}$ is continuous on $[a,b]$, differentiable on (a,b), and $f(a) = f(b)$, then $f'(c) = 0$ for some $c \in (a,b)$.

Proof. If f is constant, any c works. Otherwise f attains a maximum or minimum (Corollary 3.9) at a point $c \in (a,b)$ where $f(c) \neq f(a)$, and $f'(c) = 0$. ∎

Theorem 4.6 (4.2.2, Cauchy mean value theorem)

If $f, g : [a,b] \to \mathbb{R}$ are continuous on $[a,b]$ and differentiable on (a,b), there is $c \in (a,b)$ with $f'(c)[g(b) - g(a)] = g'(c)[f(b) - f(a)]$.

Proof. Apply Rolle to $h(x) = f(x)[g(b) - g(a)] - g(x)[f(b) - f(a)]$, for which $h(a) = h(b) = f(a)g(b) - g(a)f(b)$. ∎

Corollary 4.7 (4.2.3, mean value theorem)

If f is continuous on $[a,b]$ and differentiable on (a,b), then $f(b) - f(a) = f'(c)(b-a)$ for some $c \in (a,b)$.

Proof. Take $g(x) = x$. ∎

Corollary 4.8 (4.2.4)

For differentiable $f : (a,b) \to \mathbb{R}$: f is increasing iff $f' \geq 0$; constant iff $f' = 0$; decreasing iff $f' \leq 0$. If $f' > 0$, f is strictly increasing.

Proof. If $f' \geq 0$ and $x < y$, $f(y) - f(x) = f'(c)(y - x) \geq 0$. Conversely, if f increases, its difference quotients are ≥ 0, so $f' \geq 0$. ∎

Theorem 4.9 (4.2.5, L'Hôpital's rule)

Let f, g be differentiable near x_0 (except possibly at x_0) with $g' \neq 0$ there, $\lim_{x\to x_0} f(x) = \lim_{x\to x_0} g(x) = 0$, and $\lim_{x\to x_0} f'(x)/g'(x) = L$. Then $\lim_{x\to x_0} f(x)/g(x) = L$. The same holds for one-sided limits, for $x \to \pm\infty$, and when $g(x) \to \pm\infty$.

Proof. Set $f(x_0) = g(x_0) = 0$, making f, g continuous at x_0. For x near x_0, $g(x) \neq 0$ (by Rolle, as $g' \neq 0$), and Theorem 4.6 on the interval between x_0 and x gives $f(x)/g(x) = f'(c_x)/g'(c_x)$ with c_x between x_0 and x; as $x \to x_0$, $c_x \to x_0$. For $x \to \infty$, substitute $x = 1/t$. If $g \to \infty$ as $x \to x_0^+$: given ϵ, choose δ with $|f'/g' - L| < \epsilon$ on $(x_0, x_0 + \delta)$ and fix y there. For $x_0 < x < y$, $\frac{f(x)-f(y)}{g(x)-g(y)} = \frac{f'(c)}{g'(c)}$, and

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(y)}{g(x) - g(y)} \cdot \frac{1 - g(y)/g(x)}{1} + \frac{f(y)}{g(x)},$$

where $g(y)/g(x) \to 0$ and $f(y)/g(x) \to 0$ as $x \to x_0^+$; so $\limsup |f/g - L| \leq \epsilon$. ■

Theorem 4.10 (4.2.6, Darboux)

If $F : [a, b] \to \mathbb{R}$ is differentiable and $F'(a) < \alpha < F'(b)$, then $F'(c) = \alpha$ for some $c \in (a, b)$.

Proof. $G(x) = F(x) - \alpha x$ has $G'(a) < 0 < G'(b)$, so $G(x) < G(a)$ for x slightly larger than a and $G(x) < G(b)$ for x slightly smaller than b. Hence the minimum of G on $[a, b]$ is attained at an interior point c, where $G'(c) = 0$. ■

Corollary 4.11 (4.2.7)

If $f : I \to \mathbb{R}$ has a primitive on the interval I, then $f(J)$ is an interval for every interval $J \subset I$.

Proof. Apply Theorem 4.10 to the primitive on subintervals of J. ■

So a function with a jump discontinuity, such as the sign function, has no primitive; derivatives can be discontinuous (Example 4.1) but have the intermediate value property.

4.3 Taylor's theorem

Motivation. Polynomials are the functions we can compute with. Taylor's theorem says how well a smooth function is approximated by its Taylor polynomial and gives an explicit error term, which is the basis of local extremum tests and numerical approximations.

For f with n derivatives at a, let $T_n^f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x - a)^k$ be its Taylor polynomial at a.

Theorem 4.12 (4.3.1, generalized mean value theorem)

Let I be the closed interval with end points $a \neq b$, and $f, g : I \to \mathbb{R}$ with $f^{(n)}, g^{(n)}$ continuous on I and differentiable on $\operatorname{Int} I$. For each $x \in I$, $x \neq a$, there is c strictly between a and x with

$$[f(x) - T_n^f(x)]g^{(n+1)}(c) = [g(x) - T_n^g(x)]f^{(n+1)}(c).$$

Proof. Fix x and let $F(t) = \sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}(x - t)^k$, $G(t)$ likewise for g, for t between a and x. In $F'(t)$ the terms telescope:

$$F'(t) = \sum_{k=0}^{n} \frac{f^{(k+1)}(t)}{k!}(x - t)^k - \sum_{k=1}^{n} \frac{f^{(k)}(t)}{(k-1)!}(x - t)^{k-1} = \frac{f^{(n+1)}(t)}{n!}(x - t)^n,$$

and similarly for G. Also $F(x) = f(x)$ and $F(a) = T_n^f(x)$. Theorem 4.6 gives c with $[F(x) - F(a)]G'(c) = [G(x) - G(a)]F'(c)$; cancel the nonzero factor $(x-c)^n/n!$. ■

Corollary 4.13 (4.3.2, Taylor's theorem)

Under the hypotheses on f, for each $x \in I$ there is c between a and x with

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Proof. Apply Theorem 4.12 with $g(t) = (t-a)^{n+1}$: $T_n^g = 0$ and $g^{(n+1)} = (n+1)!$. ■

The last term is the **remainder**. If f is C^∞, the **Taylor series** of f at a is $\sum_k \frac{f^{(k)}(a)}{k!}(x-a)^k$; it may fail to converge to f: $f(x) = e^{-1/x^2}$, $f(0) = 0$, has all derivatives 0 at 0.

Proposition 4.14 (4.3.3)

Let f be C^∞ on I, $a, x \in I$, and J the closed interval between them. If $\{|f^{(n)}(t)| : t \in J, n \geq 0\}$ is bounded, the Taylor series of f at a converges to $f(x)$.

Proof. With M a bound, the remainder is at most $M|x-a|^{n+1}/(n+1)! \to 0$ (the terms of the convergent series $\sum r^n/n!$ tend to 0). ■

Example 4.2

e^x, $\sin x$, $\cos x$ satisfy the hypothesis on every bounded interval, so $e^x = \sum x^n/n!$, $\sin x = \sum(-1)^n x^{2n+1}/(2n+1)!$, $\cos x = \sum(-1)^n x^{2n}/(2n)!$ for all real x. Taylor's theorem also gives the second derivative test: if $f'(a) = 0$ and $f''(a) > 0$, f has a strict local minimum at a.

4.4 Selected exercises

Motivation. Four problems from the book's exercise set. They show the mean value theorem, Rolle's lemma and L'Hôpital's rule at work on questions that are not immediate from the definitions.

Exercise 4.4.8 (A Hölder condition of order > 1)

Let I be an interval, $M > 0$, $\alpha > 1$, and $f : I \to \mathbb{R}$ with $|f(x) - f(y)| \leq M|x-y|^\alpha$ for all $x, y \in I$. Show that f is constant. (The book's case is $M = 1$, $\alpha = 2$, $I = \mathbb{R}$.)

Solution.

Step 1. For $x \in I$ and $y \in I$, $y \neq x$,

$$\left|\frac{f(y) - f(x)}{y - x}\right| \leq M|y-x|^{\alpha-1} \longrightarrow 0 \quad (y \to x),$$

since $\alpha - 1 > 0$. So f is differentiable on I with $f' \equiv 0$ (one-sided at end points).

Step 2. Let $a < b$ in I. f is continuous on $[a,b]$ (Proposition 4.2) and differentiable on (a,b), so by Corollary 4.7, $f(b) - f(a) = f'(c)(b-a) = 0$. Hence f is constant.

For $\alpha = 1$ the conclusion fails ($f(x) = x$), and for $\alpha < 1$ as well ($f(x) = \sqrt{x}$ on $[0,\infty)$ satisfies the condition with $\alpha = \frac{1}{2}$, $M = 1$, by Example 3.3). ■

Exercise 4.4.9 (Limits of the derivative)

Let I be an open interval, $c \in I$, and $f : I \to \mathbb{R}$ continuous at c and differentiable on $I - \{c\}$. Show that if $\lim_{x\to c} f'(x) = K$ exists, then f is differentiable at c and $f'(c) = K$.

Solution.

Step 1. Let $\epsilon > 0$. Choose $\eta > 0$ with $[c-\eta, c+\eta] \subset I$ and $|f'(z) - K| < \epsilon$ for $0 < |z - c| < \eta$.

Step 2. Let $0 < h < \eta$. f is continuous on $[c, c+h]$ (at c by hypothesis, elsewhere by Proposition 4.2) and differentiable on $(c, c+h)$. By Corollary 4.7 there is $z \in (c, c+h)$ with

$$\frac{f(c+h) - f(c)}{h} = f'(z), \qquad \text{so} \qquad \left|\frac{f(c+h) - f(c)}{h} - K\right| < \epsilon.$$

Step 3. The same argument on $[c+h, c]$ for $-\eta < h < 0$ gives the same bound. Hence the difference quotient tends to K, i.e. $f'(c) = K$.

Consequently a derivative has no removable or jump discontinuities: if f' has one-sided limits at c, they equal $f'(c)$. A derivative can still be discontinuous, as Example 4.1 shows, but only because $\lim_{x\to c} f'(x)$ fails to exist. ■

Exercise 4.4.12 (Roots of $\frac{d^n}{dx^n}(x^2 - x)^n$)

Let $n \geq 1$ and $f(x) = (x^2 - x)^n = x^n(x-1)^n$. Show that $f^{(n)}$ has exactly n distinct roots, all in $(0, 1)$.

Solution.

Step 1. For $0 \leq k \leq n$, $f^{(k)}(x) = x^{n-k}h_k(x)$ for some polynomial h_k. This holds for $k = 0$ with $h_0 = (x-1)^n$; if it holds for $k < n$, then with $m = n - k \geq 1$,

$$f^{(k+1)}(x) = mx^{m-1}h_k(x) + x^m h_k'(x) = x^{m-1}(mh_k(x) + xh_k'(x)).$$

Hence $f^{(k)}(0) = 0$ for $0 \leq k \leq n-1$. Writing $f(x) = (x-1)^n x^n$, the same argument with $x - 1$ in place of x gives $f^{(k)}(1) = 0$ for $0 \leq k \leq n-1$.

Step 2. Claim: for $1 \leq k \leq n$, $f^{(k)}$ has at least k distinct roots in $(0, 1)$. For $k = 1$ apply Rolle's lemma (Lemma 4.5) to f on $[0, 1]$. If $k \leq n-1$ and $0 < \alpha_1 < \cdots < \alpha_k < 1$ are roots of $f^{(k)}$, then by Step 1 $f^{(k)}$ vanishes at the $k+2$ points

$$0 < \alpha_1 < \cdots < \alpha_k < 1.$$

Rolle's lemma on each of the $k+1$ intervals between consecutive points gives a root of $f^{(k+1)}$ inside each; these open intervals are disjoint, so the $k+1$ roots are distinct.

Step 3. f has degree $2n$ with leading coefficient 1, so $f^{(n)}$ has degree n with leading coefficient $(2n)!/n! \neq 0$. A nonzero polynomial of degree n has at most n roots, so by Step 2 $f^{(n)}$ has exactly n distinct roots, all in $(0, 1)$.

Up to a constant and the change of variable $x = (1+t)/2$, $f^{(n)}$ is the n-th Legendre polynomial. ■

Exercise 4.4.13 (Symmetric second difference quotient)

(a) Let f be differentiable on an open interval $I \ni c$ and suppose $f''(c)$ exists. Show that

$$\lim_{h\to 0} \frac{f(c+h) + f(c-h) - 2f(c)}{h^2} = f''(c).$$

(b) Show that the limit in (a) may exist although $f''(c)$ does not.

Solution.

(a)

Step 1. For small $h \neq 0$ let $N(h) = f(c+h) + f(c-h) - 2f(c)$ and $D(h) = h^2$. Both are differentiable in h, $N(h) \to 0$ by continuity of f at c, $D(h) \to 0$, and $D'(h) = 2h \neq 0$.

Step 2. We have

$$\frac{N'(h)}{D'(h)} = \frac{f'(c+h) - f'(c-h)}{2h} = \frac{1}{2}\Big(\frac{f'(c+h) - f'(c)}{h} + \frac{f'(c-h) - f'(c)}{-h}\Big)$$
$$\longrightarrow \frac{1}{2}(f''(c) + f''(c)) = f''(c).$$

Step 3. By L'Hôpital's rule (Theorem 4.9), $N(h)/D(h) \to f''(c)$.

(b) Let $f(x) = x|x|$ and $c = 0$. Then $N(h) = h|h| - h|h| - 0 = 0$, so the limit is 0. But $f'(x) = 2|x|$ for all x (at 0: $f(h)/h = |h| \to 0$), and $2|x|$ is not differentiable at 0; so $f''(0)$ does not exist. ■